StreetCare AI — logical architecture

What the system does, independent of technology: who uses it, the controls every request passes, the use cases, the rules and AI agents behind them, and the information they keep

Logical view — technology is shown in the technical and deployment architectures

1 Actors

Residents
outside the council
report street defects through Website · StreetCare app · Contact centre → the open queue

Street Care Coordinator
C · triage
scans the queue, studies reports, opens an AI-prepared review audits reviews and AI calls

Highways Inspector
I · 2 per district
decides reviews for own district; never one they opened (four-eyes) also opens reviews and audits

Customer Services Adviser
A · residents' line
answers residents: report status and documented rules only; never a likelihood or crew hours

uses pages

2 Presentation

Sign in — *everyone* — C I A
Report queue — *rank · filter by type* — C I
Report detail — *screening · evidence · estimates* — C I
Decision queue — *reviews awaiting my district* — I
Audit trail — *reviews · AI calls · chat* — C I
Street Care chat — *ask the documents / records* — C I A

every request

3 Access & control

`401` **Identity** — signed in, session live
`403` **Permission** — role allows this operation
`422` **Validation** — request well formed
`404` **Existence** — report / review is known
`403` **District** — inspector appointed there
`409` **Case rules** — one review per revision, one decision, decider ≠ opener, decision open for the assessment

allowed only

4 Use cases

Sign in / out — start or end a session pages + permissions follow from the role — *who* C I A
Browse the queue — 40 open reports, ranked by rules; filter by defect type; open one report — *who* C I
Study a report — standards screening, street works, evidence, two estimates — no AI — *who* C I
Ask Street Care — classify the question; answer from documents or records, by role — *who* C I A
Audit — reviews, decisions, AI calls with tokens and latency, chat turns — *who* C I
Open a review — 5 AI analyses on the facts → recommendation → pause for a decision — *who* C I
Decide a review — checks, then resume: documents for the decision → recorded — *who* I

applies rules

5 Rules & AI agents

Business rules — deterministic, explainable, testable

Report rules — references, units, spellings, latest revision
Standards screening — each requirement: Met / Not met / Not documented
Street-works check — live and conflicting permits, coordination triggers
Evidence analysis — sentence types; where the documents disagree
Outcome estimates — repeat likelihood + band, crew hours, overrun
Knowledge retrieval — standard sections and passages, for citation
Decision rules — decisions open, district, four-eyes, once only
Disclosure rules — what each role may hear; street figures ≥ 5 reports

facts

AI agents — interpret the facts, compute nothing

one task each · fixed labelled answer fields · every call logged

1 **Planner** → 2 **Requirements analyst** → 3 **Evidence analyst** → 4 **Risk analyst** → 5 **Review writer**

Human decision gate — the review waits, durably, for an inspector
6 **Order writer** — *ORDER · INSPECT*
7 **Resident-update writer** — *ORDER only*
Street Care assistant — *chat · 1 call*

reads · writes

6 Information

Open reports — reports, attached documents (the live queue)
Streets & works — districts, streets, street-works permits
Policy knowledge — 16 standards, guides and policies, in sections
History & models — past reports and repairs; 3 trained models
Review case files — review, recommendation, decision, documents
Paused reviews — the state each review waits in at the gate
Audit log — every AI call; every chat question and answer
Staff & sessions — 16 accounts, roles, districts, sessions

prompts

Language-model service — external
external · one call per agent task
writes prose from given facts; never decides, never computes

Decision outcomes

what each inspector decision produces

ORDER REPAIR
works order → the council's repair contractor + resident update
2 AI documents

INSPECT ON SITE
inspection request → the inspector who will visit the defect
1 AI document

REFER FOR COORDINATION
coordination brief → the network coordination team assembled by rules
0 AI documents

resident update: plain words, no staff name

The main flow — one report, from queue to resident

1 **C** opens a review → 2 **rules** compute the facts → 3 **agents 1–5** analyse, recommend → 4 **gate** review waits → 5 **I** decides (checked) → 6 **agents 6–7** draft documents → 7 **record** outcome + audit → 8 **A** tells the resident

Actor · Presentation · Rule-based · AI-assisted · Human decision / case record · Prediction & history · Information · C Coordinator · I Inspector · A Adviser